Canonical distinguishing colorings of cyclic tournaments

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Abstract

Albertson and Collins conjectured that every cyclic tournament on $\mathbb{Z}_{2q+1}$ is distinguished by coloring $0, \ldots, q$ with one color and the rest with another. I proved the conjecture with an equivalent statement regarding odd-order permutation groups. If $G \leq \mathrm{Sym}(\mathbb{Z}_n)$ has odd order and contains translation by one, then the subset $0, \ldots, (n-1)/2$ has a trivial setwise stabilizer in G. The proof makes use of an induction hypothesis on invariant block systems along with the Burnside-Schur theorem. In the primitive case, the group theorem along with the intersection sizes of translates of the interval determine the affine stabilizer. The conjecture is formally verified in Lean.

Keywords: cyclic tournament; distinguishing coloring; regular subset; permutation group.
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1 Introduction

A coloring of a digraph is *distinguishing* if the identity is the only automorphism that fixes every color class. A *cyclic tournament* $T(n; S)$ is a digraph on vertex set $\mathbb{Z}_n$ with an arc $i \to j$ whenever $j - i \in S$, where $S \subseteq \mathbb{Z}_n \setminus 0$ contains exactly one element of each pair $d, -d$. Translation by one is an automorphism of $T(n; S)$. The conjecture is for the case where $n = 2q + 1 \geq 3$. Albertson and Collins [1, Conjecture 2.4] posed the conjecture in terms of two-colorings. Meslem and Sopena [6, Conjecture 2] stated the conjecture as follows. The first color class is $I_n = 0, \ldots, q$.

The original version uses the set $0, ..., q - 1$. These statements are equivalent: the complement of the last set is $I_n + q$, a set and its complement have the same setwise stabilizer, and translation conjugates stabilizers. Meslem and Sopena [6, Conjecture 2] established the prescribed coloring for several classes of tournaments including rigid Paley tournaments and rigid cyclic tournaments, for which one of the canonically induced subtournaments is rigid.

A subset is said to be *regular* for a permutation group, if its setwise stabilizer is trivial. Gluck's Regular Subset Theorem [5] guarantees the existence of a distinguishing labeling with at most two labels, for every finite tournament. This is also shown in Meslem and Sopena [7, Section 3, Theorem 8].[1]

In this paper, we investigate if the prescribed interval works for every connector set S.

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[1]For primitive permutation groups not containing the alternating group, Cameron, Neumann, and Saxl [3] prove existence of a regular subset apart from finitely many exceptions. This conclusion does not specify an interval in given cyclic coordinates.

I prove the same for all connector sets S. I formulate the result in terms of odd-order permutation groups containing the standard cyclic shift. This formulation is equivalent to the tournament statement:

Theorem 1.1. *Let n be a positive odd integer. If $G \leq \mathrm{Sym}(\mathbb{Z}_n)$ has odd order and contains $\tau : x \mapsto x+1$, then $I_n = \{0, \ldots, (n-1)/2\}$ is regular for G.*

The proof makes use of strong induction on n. In an imprimitive action, the two interval sizes within the blocks determine a canonical half of the block indices. The induction first fixes blocks and then fixes their vertices. In a primitive action, Burnside-Schur reduces the problem to an affine action of prime degree. The next lemma determines the possible affine stabilizers by counting intersections of interval translates. Section 4 gives the tournament proof and establishes the converse.

2 The affine interval lemma

Lemma 2.1. *Let $p = 2q+1$ be an odd prime and let $I = \{0, \ldots, q\} \subseteq \mathbb{F}_p$. The setwise stabilizer of I in $\mathrm{AGL}_1(p)$ consists of the identity and the reflection $x \mapsto q - x$.*

Proof. For $d \in \mathbb{F}_p$, write $C(d) = |I \cap (I+d)|$. Then $C(0) = q+1$, and, for a nonzero d represented in $\{1, \ldots, 2q\}$,

$$C(d) = q + 1 - \min\{d, p-d\}. \tag{1}$$

Indeed, if $1 \leq d \leq q$, the intersection is $\{d, \ldots, q\}$. The other values follow from $C(-d) = C(d)$, since translating $I \cap (I+d)$ by $-d$ gives $(I-d) \cap I$. Thus the largest intersection at a nonzero shift has size q, and it occurs exactly at $d = 1$ and $d = -1$.

Suppose $f(x) = \alpha x + \beta$, with $\alpha \neq 0$, preserves I. The identity $f(I+d) = I + \alpha d$ and bijectivity imply $C(\alpha d) = C(d)$. Setting $d = 1$ gives $\alpha \in \{1, -1\}$. If $\alpha = 1$, preservation of I gives $I + \beta = I$. Equation (1) excludes a nonzero β, so f is the identity. Finally, the reflection $r(x) = q - x$ preserves I. If $\alpha = -1$, then $r \circ f$ is a translation preserving I, and hence is the identity. Thus $f = r$. □

3 Proof of the group theorem

Proof of Theorem 1.1. I use strong induction on the positive odd integer n. For $n = 1$, the assertion is immediate. Two elementary facts will be used in the inductive step. A subset and its complement have the same setwise stabilizer. Also, when a group contains all translations, regularity of a subset implies regularity of each of its translates, by conjugating a stabilizer with the corresponding translation. Since $\tau \in G$, the action is transitive and G contains every translation of $\mathbb{Z}_n$.

Suppose first that the action is imprimitive, and let B be a nontrivial block containing 0. For each $t \in B$, the point t belongs to $B \cap (B+t)$, so the block property gives $B + t = B$. It follows that B is an additive subgroup: it is closed under addition, and $0 \in B + t$ implies $-t \in B$. Its translates form the block system. Therefore, for $n = ab$ with $1 < a, b < n$, the blocks are

$$B_r = \{r + at : 0 \leq t < b\}, \qquad 0 \leq r < a.$$

Both factors are odd. Write $a = 2u+1$ and $b = 2v+1$. Since $(n-1)/2 = av + u$,

$$I_n \cap B_r = \begin{cases} \{r + at : 0 \leq t \leq v\}, & 0 \leq r \leq u, \\ \{r + at : 0 \leq t < v\}, & u < r < a. \end{cases} \tag{2}$$

The representatives satisfy $0 \le r + at < ab$, so this calculation involves no wraparound.

Let $g \in G$ preserve I_n. Its induced permutation on the blocks preserves their intersection sizes with I_n. The blocks with the larger intersection size $v+1$ are indexed precisely by I_a. The induced permutation group on block indices has odd order, as an image of G, and contains $r \mapsto r+1$ on $\mathbb{Z}_a$. By induction, the induced permutation of g is the identity. Thus g fixes every block setwise.

For each r, consider the group induced on B_r by its setwise stabilizer in G. This group has odd order, as an image of a subgroup of G, and contains the restriction of τ^a. In the coordinate $r + at \leftrightarrow t \in \mathbb{Z}_b$, that restriction is translation by one. If $r \le u$, the preserved intersection in (2) is I_b. If $r > u$, its complement in B_r corresponds to $\{v, \dots, 2v\} = I_b + v$. In both cases, induction and the two elementary facts above make the restriction of g the identity. Hence g fixes every vertex.

It remains to consider a primitive action. The Burnside–Schur theorem [4, opening theorem] states that a primitive finite permutation group containing a regular cyclic subgroup is either 2-transitive or permutationally isomorphic to a subgroup of $\mathrm{AGL}_1(p)$ in prime degree p. A 2-transitive group of degree $n \ge 3$ has order divisible by $n(n-1)$, which is even. Hence, p is prime, and the affine case holds. The affine case holds in the original residue coordinates to make the interval argument valid.

Let $f : \mathbb{Z}_p \to \mathbb{F}_p$ be a bijection that conjugates G to a subgroup of $\mathrm{AGL}_1(p)$. The order of $f\tau f^{-1}$ is p. An affine map of order p whose multiplier is not 1 is conjugate to multiplication by that number, and has a fixed point. Hence, an affine map of order p has a fixed point.

It is clear that $f\tau f^{-1}$ is a non-zero translation. Thus $f(j+1) = f(j) + c$ and $f(j) = f(0) + jc$ for all $j \in \mathbb{Z}_p$. Composing f with $z \mapsto (z - f(0))/c$ gives the original residue identification. Since the last change of variables is affine, every element of G is affine in those variables. Therefore, by Lemma 2.1, an element of G preserving I_p is the identity, or the non-identity involution, $x \mapsto q - x$. This completes the proof. □

4 Cyclic tournaments

Corollary 4.1. *For every cyclic tournament $T(2q+1; S)$ with $q \ge 1$, the coloring*

$$\lambda(i) = \begin{cases} 1, & 0 \le i \le q, \\ 2, & q < i \le 2q \end{cases}$$

is distinguishing.

Proof. The automorphism group of a finite tournament has odd order. Otherwise, by Cauchy's theorem, there would exist a non-identity involution. Such a permutation would interchange two distinct vertices and reverse the unique arc between them, so it cannot be an automorphism. Since τ also belongs to $\mathrm{Aut}(T)$, by Theorem 1.1, I_{2q+1} is regular in $\mathrm{Aut}(T)$. This is exactly the color preserving statement needed for the corollary. For easier reference, the constants in Corollary 4.1 are relative to the tournament $T(2q+1; S)$ given in the statement. □

The group formulation is also a consequence of the tournament statement. To prove the converse for $n > 1$, let $G \le \mathrm{Sym}(\mathbb{Z}_n)$ have odd order and contain τ. No element of G can interchange two distinct vertices: an element that does so has a 2-cycle and therefore has even order. Hence no orbit of G on ordered pairs of distinct vertices meets its reversal. Reversal partitions these orbits into pairs. Choose one orbit from each pair and let their union be the arc set of a tournament T. Then $G \le \mathrm{Aut}(T)$. Since $\tau \in G$, the arc set is translation invariant, so T is a cyclic tournament in the original residue coordinates. Corollary 4.1, viewed as the prescribed-coloring assertion for all cyclic

tournaments, makes I_n regular for $\mathrm{Aut}(T)$ and therefore for G. The case $n = 1$ is immediate. Thus the group theorem and the tournament conjecture are equivalent.

5 Formal verification

I have formalized the group theorem and the tournament conclusion in Lean, using proof terms for the Burnside–Schur step from AxiomMath's formalization [2]. The completed proof is recorded at commit `2802921` in the repository

`https://github.com/alexgocardinal/albertson-collins`.

The proof was built with Lean 4.35.0-rc3 and a pinned Mathlib revision. The final axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`; the classification theorem is proved rather than left as an additional hypothesis. Appendix A gives the theorem signatures, their relation to the mathematical statements, the exact classification proposition, and commands for reproducing the build. The finite computational checks in the repository are not premises of the formal proof.

Data availability. The proof does not use any research datasets. The repository above contains the Lean sources, dependency pins, computational scripts, and example graphs.

AI assistance. ChatGPT/Codex (OpenAI) proof read the supplied proof, implemented finite checks, developed and debugged the Lean formalization, and edited the exposition.

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A Formal statements and reproduction

A.1 Specification of the interval and stabilizer

The declarations below are in the namespace `AlbertsonCollins`. The universe parameter `u` is arbitrary. The relevant definitions are:

```
def PermStabilizes {V : Type u} (S : Set V)
    (f : Equiv.Perm V) : Prop :=
  ∀ x, f x ∈ S ↔ x ∈ S

def RegularSubset {V : Type u}
    (G : Subgroup (Equiv.Perm V)) (S : Set V) : Prop :=
  ∀ f, f ∈ G → PermStabilizes S f → f = 1

def halfSet (n : Nat) : Set (Fin n) :=
  {x | x.val ≤ (n - 1) / 2}

def CanonicalHalfGroupStatement : Prop :=
  ∀ n : Nat, Odd n →
    ∀ G : Subgroup (Equiv.Perm (Fin n)),
      Odd (Nat.card G) → finRotate n ∈ G →
        RegularSubset G (halfSet n)
```

Here `Fin n` represents the labels $0, \dots, n-1$, and `finRotate n` is their cyclic successor. For positive n, the map $x \mapsto x.\texttt{val} \bmod n$ identifies these labels with $\mathbb{Z}_n$ and carries the rotation to translation by one. Oddness of n ensures positivity. The group is finite because it is a subgroup of the permutation group of `Fin n`. Thus `CanonicalHalfGroupStatement` is precisely Theorem 1.1, with the same prescribed labels and all of its quantifiers.

For a tournament `T`, the predicate and automorphism subgroup are specified by

```
def ArcPreserving {V : Type u} (T : Tournament V)
    (f : V → V) : Prop :=
  ∀ x y, T.Arc (f x) (f y) ↔ T.Arc x y
```

The subgroup `T.autGroup` consists of the permutations preserving `T.Arc`. The structure `Tournament V` requires a loopless, asymmetric arc relation with an arc in one direction between every pair of distinct vertices.

The full type signatures of the final results are as follows; only their proof terms are suppressed.

```
theorem canonicalHalfGroup_proved : CanonicalHalfGroupStatement

theorem tournament_half_regular
    (q : Nat) (T : Tournament (Fin (2 * q + 1)))
    (hcyclic : ArcPreserving T (finRotate (2 * q + 1))) :
    RegularSubset T.autGroup (halfSet (2 * q + 1))
```

For $n = 2q + 1$, membership in `halfSet n` is exactly $x \le q$. The hypothesis `hcyclic` fixes the cyclic coordinates used in the coloring. A permutation preserves the two named colors if and only if it preserves this subset setwise, so the second signature gives Corollary 4.1. Its additional case $q = 0$ is the trivial one-vertex case.

A.2 The classification boundary

The local classification proposition is displayed in full below. Here `residueTranslation t` is $x \mapsto x + t$ on `ZMod n`. The predicate `MulAction.IsBlock G B` is the usual block condition for the action of G. The negated existential says exactly that no proper block containing zero contains a second vertex. The presence of translation by one makes the action transitive, so this is the primitive case.

```
def BurnsideSchurStatement : Prop :=
  ∀ n : Nat, 1 < n →
    ∀ G : Subgroup (Equiv.Perm (ZMod n)),
      residueTranslation (1 : ZMod n) ∈ G →
      (¬ ∃ B : Set (ZMod n),
        (0 : ZMod n) ∈ B ∧
        MulAction.IsBlock G B ∧
        B ≠ Set.univ ∧
        ∃ x : ZMod n, x ∈ B ∧ x ≠ 0) →
      (∀ x y u v : ZMod n, x ≠ y → u ≠ v →
        ∃ g : G, g.1 x = u ∧ g.1 y = v) ∨
      (Nat.Prime n ∧
        ∃ e : Equiv.Perm (ZMod n),
          ∀ f : Equiv.Perm (ZMod n), f ∈ G →
            ∃ a b : ZMod n,
              ∀ z : ZMod n, e (f (e.symm z)) = a * z + b)
```

```
theorem burnsideSchurStatement_proved : BurnsideSchurStatement
```

The first alternative is 2-transitivity; the second gives affine coordinates in prime degree. In the latter, the multiplier is necessarily nonzero because the displayed map is a permutation. The proof of `burnsideSchurStatement_proved` calls the following composite-degree and prime-degree declarations:

```
CFSG.BurnsideSchur.isMultiplyPretransitive_of_not_prime
CFSG.BurnsideSchur.exists_affine_of_not_isMultiplyPretransitive
```

The former proves 2-transitivity of a primitive action on a finite cyclic additive group of nonprime cardinality containing every translation. The latter proves that a permutation in a group on $\mathbb{Z}_p$ containing every translation is affine whenever the group is not 2-transitive. The bridge uses these proved results to supply the displayed proposition and then the group and tournament theorems.

A.3 Source versions and build commands

The proof checkpoint is `2802921`, with Lean 4.35.0-rc3. The pinned Mathlib and AxiomMath/CFSG source revisions are:

```
Mathlib: b63f6e8a68d220e3b3bc4f3792bb53650d375f24
CFSG:    69cb972f587a3ecc035e092253167ff2e303c962
```

The classification source directory is `lean/CFSG/Permutation/BurnsideSchur`; attribution and licensing are recorded in `lean/CFSG/UPSTREAM.json` and `lean/CFSG/LICENSE`.

From a fresh checkout, the following commands build the pinned proof and print its final signatures and axiom reports:

```
git clone https://github.com/alexgocardinal/albertson-collins.git
cd albertson-collins
git checkout --detach 2802921
cd lean
lake exe cache get Mathlib
lake build
lake env lean AlbertsonCollins/BurnsideSchurBridge.lean
```

The final command prints the three axiom reports and the theorem signatures.